DRAFT BOX

Match Engine & Tactics Rulebook

How teams actually play against each other — every rating, formula, and formation matchup used by the match simulator, laid out exactly as implemented.

Draft Box V 1.1.5 — github.com/DeepInkGroup/draft-box

1. Overview

Every match in Draft Box — bot vs. bot, human vs. bot, or human vs. human — is decided by the same engine. It has four ingredients:

- **Player quality.** The overall rating of each of the 11 drafted (or bot-picked) players, weighted by how advanced or deep their exact slot sits on the pitch, and by an exponential quality curve for who actually gets on the scoresheet (Section 4, Section 9).
- **Goalkeeper quality.** The opposing keeper's own overall rating — not just their contribution to the averaged Defense number — separately suppresses or inflates the shooting side's expected goals, in open play and in a penalty shootout alike (Section 4.1).
- **Formation shape.** A tactical fingerprint derived purely from the formation's slot layout — how attack-minded it is, how defensively solid it is, and how wide it stretches the pitch.
- **Chance.** Goals are sampled from a Poisson distribution around each team's "expected goals" number, so the better team wins more often — not every time.

Nothing here is hardcoded per formation or per team — every number below is derived from the same slot coordinates used to draw the pitch diagram in the app, so the numbers in this document are exactly what the live server computes.

2. The 16 Formations

Every formation is an explicit 11-slot layout (not just a GK/DF/MF/FW count) — this is what lets a player choose exactly which side of defense, midfield, or attack a drafted player fills.

Formation	GK	DF	MF	FW	Style
4-3-3	1	4	3	3	Attacking with width. Three forwards create constant threat.
4-4-2	1	4	4	2	Balanced and reliable. Two banks of four with a strike partnership.
4-2-3-1	1	4	5	1	Defensive solidity with a creative No.10 behind a lone striker.
4-5-1	1	4	5	1	Ultra-compact midfield. Absorb pressure and hit on the counter.
3-4-3	1	3	4	3	High-risk, high-reward. Bold width in attack, exposed at the back.
3-5-2	1	3	5	2	Wing-backs provide width while a five-man midfield dominates the center.
5-4-1	1	5	4	1	Fortress at the back. Sit deep, stay compact, strike on the break.
4-1-2-1-2	1	4	4	2	A narrow diamond in midfield builds through the center.
4-4-1-1	1	4	4	2	A second striker links midfield and attack just behind the target man.
5-3-2	1	5	3	2	Three at the back become five in defense. Cautious, hard to break down.
3-4-1-2	1	3	5	2	A free-roaming No.10 supports a strike duo behind a solid back three.
4-2-2-2	1	4	4	2	Two holding midfielders shield the back four while two playmakers link the strikers.
5-2-3	1	5	2	3	A back five with only two in the middle — firepower up top, engine room can get overrun.

Formation	GK	DF	MF	FW	Style
5-3-1-1	1	5	3	2	Extreme defensive solidity, a target man and support striker linking play.
3-2-4-1	1	3	6	1	A back three with a double pivot, exploding into four attacking outlets.
4-2-4	1	4	2	4	Four forwards, two midfielders. Direct but leaves acres of space to exploit.

3. Formation Shape (the tactical fingerprint)

Three numbers are derived once per formation, straight from its slot list — no per-formation tuning:

```
defShape = SUM(every DF slot) + SUM(every MF slot with y >= 55, i.e. sitting deep)
atkShape = 1.5 × SUM(every FW slot) + SUM(every MF slot with y <= 35, i.e. pushed forward)
width = (max x) - (min x) among outfield slots
```

(y is pitch depth: 0 = the opponent's goal line, 100 = your own goal / GK. x is 0-100 left-to-right.) A formation with a back five and only one striker (5-4-1) has very high defShape and very low atkShape. A front four with two holding midfielders (4-2-4) is the exact opposite.

Formation	atkShape	defShape	width
4-3-3	4.5	4	70
4-4-2	3.0	4	70
4-2-3-1	4.5	6	70
4-5-1	1.5	5	76
3-4-3	4.5	3	72
3-5-2	3.0	4	80
5-4-1	1.5	5	80
4-1-2-1-2	4.0	5	70
4-4-1-1	3.0	4	70
5-3-2	3.0	5	80
3-4-1-2	4.0	3	72
4-2-2-2	5.0	6	70
5-2-3	4.5	5	80
5-3-1-1	3.0	6	80
3-2-4-1	5.5	5	76
4-2-4	6.0	4	70

4. Player Ratings

Every filled slot gets a continuous attack/defense weight from its own pitch depth (y):

```
attackWeight(slot) = 1 - y / 100
defenseWeight(slot) = y / 100
```

A striker (y~14) is weighted ~0.86 toward attack and only ~0.14 toward defense. A center back (y~78) is the reverse. A team's two headline numbers are then a weighted average of its 11 players' overall ratings:

```
TeamAttack = SUM(overall_i × attackWeight_i) / SUM(attackWeight_i) + starBonus
TeamDefense = SUM(overall_i × defenseWeight_i) / SUM(defenseWeight_i) + starBonus
```

Star Bonus

starBonus = min(3, numRealStarPlayersOnTheXI) × 1.5, added to both Attack and Defense. A squad with three or more of the handful of real, well-known stars seeded into the data gets a flat +4.5 rating lift, reflecting composure and quality that a raw overall average under-counts. A fourth star adds nothing further — the bonus caps out.

Captain Bonus (room-optional)

When a room has captains enabled, every member picks one of their own 11 players as captain once their draft is complete — the whole room must confirm a captain (if the option is on) before the World Cup can begin. That player's overall is boosted before the weighting above:

```
effectiveOverall(captain) = min(99, overall + 6)
```

Bots never have a captain — this is a reward for the deliberate choice a human squad-builder makes.

4.1 Goalkeeper Factor

A team's goalkeeper already pulls the averaged Defense rating up or down like any back-line player — but a genuinely elite or genuinely shaky keeper has an outsized, individually-recognizable effect on real matches beyond that average. The starting keeper's own overall gets a second, separate say: it directly scales the opposing side's expected goals, in open play, extra time, and penalty kicks alike.

```
gkModifier(gkOverall) = clamp( 1 - (gkOverall - 75) × 0.008 , 0.8, 1.2 )
```

75 overall is neutral (×1.0). A genuinely elite keeper (88-92+, Alisson/Courtois territory) cuts the opponent's expected goals by up to ~12-16%; a weak one concedes proportionally more — real football's "a great keeper single-handedly wins you points" effect, distinct from the rest of the back line. Simulated: an attacking side facing a 58-overall keeper instead of a 92-overall one scored roughly 37% more goals on average, all else equal.

5. Chemistry

Beyond individual quality, how well an XI plays as a unit matters. Chemistry is three parameters, each 0-1, averaged into one score:

```
Linkage (L) = share of "adjacent" pitch partnerships sharing a real source team
Balance (B) = clamp( 1 - stdDev(overalls) / 20 , 0, 1 )
Leadership (S) = min(1, starPlayersOnXI / 3)
Chemistry = (L + B + S) / 3
multiplier = 0.94 + Chemistry × 0.12
```

The multiplier applies to both Attack and Defense, after the captain bonus. A 0.94-1.06 range means chemistry, like the formation edge, is a real but modest swing — it rewards a cohesive XI without letting it override raw quality.

5.1 Linkage — who's "adjacent"?

Every formation's slot layout defines a set of local partnerships: two slots within a short distance of each other on the pitch (a center back and the full back beside him, a central midfield trio, and so on — the same x/y coordinates used everywhere else in this document, no separate data). Linkage is the fraction of those partnerships where both players were drafted from the same real national team.

5.2 Worked example

An XI built entirely from Brazil's real squad (bots always draw their whole XI from one nation) versus a "franken-XI" of the same average overall (85) assembled from 11 different countries, both in a 4-3-3:

Squad	Linkage	Balance	Leadership	Chemistry	Multiplier
Real national squad	1.00	0.89	1.00	0.96	×1.056
11-country all-star XI	0.00	1.00	1.00	0.67	×1.020

Simulated 3,000 times at equal overall (85 each): the real national squad won 54% of the time to the all-star XI's 22% (rest drawn). This is a deliberate trade-off, not a bug: bot-controlled real countries are guaranteed perfect Linkage, so a human squad built purely by chasing the highest overalls across every country in the draft pool will often be trading chemistry away for star power — both are viable strategies.

6. Match Simulation

6.1 Formation Edge

How much team X's attacking shape overwhelms team Y's defensive shape, plus a bonus for X being meaningfully wider than Y:

```
edge(X->Y) = clamp( (atkShape_X - defShape_Y) × 0.4 + (width_X - width_Y) × 0.06 , -6, +6
)
```

Worked example (pure width, everything else equal). 4-4-2 and 3-5-2 have identical atkShape (3.0) and defShape (4) — the only difference is width (70 vs. 80). Yet:

```
edge(3-5-2 attacking -> 4-4-2 defending) = +0.20
edge(4-4-2 attacking -> 3-5-2 defending) = -1.00
```

3-5-2's extra width alone is worth a full 1.2-point swing in its favor — stretching a narrower back four creates gaps a flat 4-4-2 doesn't have to worry about defending.

6.2 Expected Goals

Base expected goals come from the Attack/Defense gap and the formation edge, then the opposing goalkeeper's own quality (Section 4.1) scales the number a second time:

```
baseXG(X vs Y) = clamp( 1.35 + (Attack_X - Defense_Y) / 18 + edge(X->Y) / 10 , 0.15, 4.5 )
xG(X vs Y) = clamp( baseXG(X vs Y) × gkModifier(GK_Y.overall) , 0.1, 5 )
```

Both teams' xG are computed independently and simultaneously (X attacking into Y's defense, and Y attacking into X's defense). The formation edge divides by 10 while the base 1.35 and player-quality term dominate the number — the tactical matchup nudges the scoreline by roughly half a goal at most, and even the goalkeeper factor tops out around ±15-20%. Player quality is still the primary driver of most results.

6.3 Goals

The actual scoreline is sampled from a Poisson distribution using each team's own xG as the mean — the standard way to turn an "expected goals" number into a real, whole-number, occasionally-surprising result.

6.4 Knockout draws → extra time → penalties

A knockout match level after 90 minutes doesn't jump straight to penalties — it plays a proper 30-minute extra time first, and only goes to a shootout if the sides are still level after that.

Extra Time

Two 15-minute periods (minutes 91-120), lower-scoring than normal time since legs are tired and sides play more cautiously — modeled as roughly a third of a full match's expected goals, still passed through the same goalkeeper factor:

```
etXG(X vs Y) = baseXG(X vs Y) × gkModifier(GK_Y.overall) × (30 / 90)
etGoals_X = Poisson( etXG(X vs Y) )
```

Penalty Shootout

If the score is still level after extra time, a real kick-by-kick shootout decides it — not an instant coin flip. Each side's best takers (by overall) go first; success chance is weighted by both the kicker's own quality and the opposing keeper's:

```
kickProb(kicker, opposingGK) = clamp( (0.62 + (kicker.overall - 75) / 200) × gkModifier(op
posingGK.overall) , 0.35, 0.92 )
```

Standard rules: 5 rounds each, with the real-football early-stop rule (the shootout ends as soon as the trailing side can no longer mathematically catch up within the remaining initial-round kicks), then sudden death — one kick each per round — until someone is left standing.

A player already sent off (Section 6.5) is excluded from taking penalties, exactly as they're excluded from scoring or assisting any goal after their dismissal — they're off the pitch, so they can't be involved in anything that happens afterward.

6.5 Cards & the Red Card Morale Hit

Yellow cards are resolved as a sequence of 0-4 incidents (drawn as floor(random × 6)) spread across the match, tracked per player as they happen — a player who's already carded and picks up a second yellow is properly sent off, rather than just being handed a second unrelated yellow event. On top of that, each side independently has a small 3.5% chance of a straight red (restricted to a player with no card yet this match, so no player's history ever contradicts itself). A dismissed player cannot receive any further card that match, and is removed from the pool of eligible scorers, assisters, and penalty takers for the remainder of the match — a sent-off player literally cannot be involved in anything that happens after their own dismissal minute.

Either way, a dismissal also rattles the sent-off player's own team for a short spell afterward, regardless of that individual player's own quality. Since the engine decides a match in one shot rather than minute-by-minute, the "short spell" is expressed as a fraction of the full 90 minutes and converted into an average-over-the-match Attack + Defense reduction, applied before the Expected Goals step (6.2):

```
affectedMinutes = 15, or 15 × 1.8 = 27 if the sent-off player is the captain
moraleImpact = 0.20 × (affectedMinutes / 90)
Attack, Defense ×= (1 - moraleImpact)
```

That works out to roughly a 3.3% temporary rating cut for an ordinary player sent off, or about 6% if the captain is the one dismissed — small individually, but it measurably lowers a red-carded side's average goals scored over many matches. Bots never have a captain (Section 4), so this only doubles up for human squads that chose one.

7. Formation Matchups

Holding player quality equal, here is every formation's best and worst matchups by pure formation edge (positive = advantage attacking into that opponent's defense). Remember: this is a nudge on top of player quality, not a guarantee — a Brazil XI in a "weak" matchup will still usually beat a Jordan XI in a "strong" one.

Formation	Strong against	Struggles against
4-3-3	3-4-3 (+0.5), 3-4-1-2 (+0.5), 4-4-2 (+0.2)	5-3-1-1 (-1.2), 5-2-3 (-0.8), 5-3-2 (-0.8)
4-4-2	3-4-3 (-0.1), 3-4-1-2 (-0.1), 4-3-3 (-0.4)	5-3-1-1 (-1.8), 5-2-3 (-1.4), 5-3-2 (-1.4)
4-2-3-1	3-4-3 (+0.5), 3-4-1-2 (+0.5), 4-3-3 (+0.2)	5-3-1-1 (-1.2), 5-2-3 (-0.8), 5-3-2 (-0.8)
4-5-1	3-4-3 (-0.4), 3-4-1-2 (-0.4), 4-3-3 (-0.6)	5-3-1-1 (-2.0), 5-2-3 (-1.6), 5-3-2 (-1.6)
3-4-3	3-4-1-2 (+0.6), 4-3-3 (+0.3), 4-4-2 (+0.3)	5-3-1-1 (-1.1), 5-2-3 (-0.7), 5-3-2 (-0.7)
3-5-2	3-4-3 (+0.5), 3-4-1-2 (+0.5), 4-3-3 (+0.2)	5-3-1-1 (-1.2), 5-2-3 (-0.8), 5-3-2 (-0.8)
5-4-1	3-4-3 (-0.1), 3-4-1-2 (-0.1), 4-3-3 (-0.4)	5-3-1-1 (-1.8), 5-2-3 (-1.4), 5-3-2 (-1.4)
4-1-2-1-2	3-4-3 (+0.3), 3-4-1-2 (+0.3), 4-3-3 (+0.0)	5-3-1-1 (-1.4), 5-2-3 (-1.0), 5-3-2 (-1.0)
4-4-1-1	3-4-3 (-0.1), 3-4-1-2 (-0.1), 4-3-3 (-0.4)	5-3-1-1 (-1.8), 5-2-3 (-1.4), 5-3-2 (-1.4)
5-3-2	3-4-3 (+0.5), 3-4-1-2 (+0.5), 4-3-3 (+0.2)	5-3-1-1 (-1.2), 5-2-3 (-0.8), 5-4-1 (-0.8)
3-4-1-2	3-4-3 (+0.4), 4-3-3 (+0.1), 4-4-2 (+0.1)	5-3-1-1 (-1.3), 5-2-3 (-0.9), 5-3-2 (-0.9)
4-2-2-2	3-4-3 (+0.7), 3-4-1-2 (+0.7), 4-3-3 (+0.4)	5-3-1-1 (-1.0), 5-2-3 (-0.6), 5-3-2 (-0.6)
5-2-3	3-4-3 (+1.1), 3-4-1-2 (+1.1), 4-3-3 (+0.8)	5-3-1-1 (-0.6), 5-3-2 (-0.2), 5-4-1 (-0.2)
5-3-1-1	3-4-3 (+0.5), 3-4-1-2 (+0.5), 4-3-3 (+0.2)	5-2-3 (-0.8), 5-3-2 (-0.8), 5-4-1 (-0.8)
3-2-4-1	3-4-3 (+1.2), 3-4-1-2 (+1.2), 4-3-3 (+1.0)	5-3-1-1 (-0.4), 5-2-3 (-0.0), 5-3-2 (-0.0)
4-2-4	3-4-3 (+1.1), 3-4-1-2 (+1.1), 4-3-3 (+0.8)	5-3-1-1 (-0.6), 5-2-3 (-0.2), 5-3-2 (-0.2)

Reading this: very defensive back-five shapes (5-3-1-1, 5-2-3, 5-3-2) are hard for anyone to break down, so they show up as the toughest matchup across the board. Wide, attack-heavy shapes (4-2-4, 3-2-4-1, 5-2-3) punish narrow or thin-on-defense formations (3-4-3, 3-4-1-2) the hardest. Bots are assigned a random formation for every tournament, so these matchups play out differently every run.

8. Tournament Format & Room Options

8.1 Full Tournament (default)

- 48 teams → 12 groups of 4, drawn at random.
- Each group plays a round robin (3 matchdays, 6 matches).
- Top 2 of each group (24) + the 8 best third-placed teams advance to the Round of 32 — the real 2026 FIFA World Cup format.
- Round of 32 → Round of 16 → Quarter-Finals → Semi-Finals → Final, single elimination.
- Any knockout draw goes to extra time, then penalties if still level (formulas in Section 6.4).

8.2 Tournament Length (room-optional)

The room creator picks one of three lengths before the draft starts. Every human drafter is guaranteed a slot in whichever bracket size is chosen; bot teams are randomly trimmed down from the full 48 to fill the rest.

Length	Starting teams	What happens
Off (default)	48	Full group stage → Round of 32 → ... → Final
Blitz	32	Skips the group stage entirely, starts directly at the Round of 32
Top 8 (1/4)	8	Skips straight to the Quarter-Finals with only 8 teams

8.3 Restrict Draft Teams (room-optional)

The room creator can optionally limit which nations are eligible to be revealed during the draft to a hand-picked list of at least 4 teams, grouped by real football confederation (UEFA, CONMEBOL, CONCACAF, AFC, CAF, OFC) in the room-creation UI for faster picking — leave it empty for the default: all 48 teams eligible. This only affects which teams appear during drafting — the World Cup simulation itself still fields all of the tournament's bot and human slots as normal. Every nation reveals independently to each drafter, so if two members of a room are both shown the same country at the same time, they each pick from that team's remaining available players on their own — one member's pick never blocks or disrupts the other's reveal.

8.4 Pick Timer (room-optional)

The room creator sets how long each drafter has to choose a player once a team is revealed to them: 10s, 20s, 30s, 45s, 60s, or No Limit (no countdown or auto-pick at all — the drafter takes as long as they need).

8.5 Starting the Tournament (multiplayer)

In a multiplayer room, the World Cup does not start the instant the last drafter finishes — every member first sees a "waiting for other players" screen while anyone is still drafting. Once everyone has completed their draft (and set a captain, if that option is on), the room creator gets a Start Tournament button and confirms it explicitly; only then does the bracket lock in and the tournament begin, with each member's championship-odds estimate showing once everyone's squad is finalized. Singleplayer rooms start automatically as soon as the lone drafter finishes, since there's no one else to wait for.

9. Live Match Reports

Every simulated match generates an event timeline — goals with scorer and (usually) an assist, plus yellow/red cards — using the same slot-weighting logic as everywhere else in this document, sharpened by an individual quality curve so a genuine star produces disproportionately more of a team's output than a squad-depth player in a similar slot, matching how real football output actually concentrates in a handful of players:

```
qualityWeight(overall) = 1.045 ^ (overall - 75)
scorerWeight(p) = (1 - slotY(p) / 100) × qualityWeight(p.overall)
assistWeight(p) = (1 - |slotY(p) - 45| / 60) × qualityWeight(p.overall)
```

75 overall is the pivot (weight 1x); roughly a 4x spread in goal-involvement odds between a 60 and a 90 overall player in an otherwise-similar slot. A player already sent off (Section 6.5) is never eligible as scorer or assist for a goal that happens after their dismissal.

Purely narrative — events don't feed back into the score, which is already decided by Section 6.

Results arrive one stage at a time: nobody is forced to watch every match in a round at once, and one person clicking through doesn't drag anyone else's screen forward. Each human's own match plays out on a live ticking clock — 90 minutes, extended to a full 120 with extra time, then a kick-by-kick penalty feed, exactly as it happened in the simulation — while every other match in that round is available as a static final report via a match picker. A pre-tournament screen also shows a championship-odds estimate and a predicted top scorer/assist for the viewer's own squad, derived from the same rating and slot-weighting formulas above.

9.1 Team Performance Stats

Every report — live or static — shows a full stat line for both sides, derived straight from the same post-chemistry, post-red-card, post-goalkeeper-factor Attack/Defense/xG numbers used for the Expected Goals step, not simulated independently:

```
qualityDiff = (Attack_A + Defense_A) - (Attack_B + Defense_B)
possession_A = clamp( 50 + qualityDiff × 0.6 , 32, 68 ); possession_B = 100 - possession_A
passAccuracy_X = clamp( 68 + (avgRating_X - 75) × 0.7 , 55, 94 ), avgRating = (Attack+Defe
nse)/2
totalPasses = 780 ± up to 60 (match tempo, random); passes_X = round(totalPasses × possess
ion_X / 100)
shotsOnTarget_X = max(goals_X, round(xG_X / 0.3)); shots_X = round(shotsOnTarget_X / 0.42)
cornersTotal = 7-13 (random); corners_X = round(cornersTotal × possession_X / 100)
fouls_X = round(clamp(10 + (possession_Y - possession_X) × 0.08 ± up to 2 , 4, 20))
yellowCards_X, redCards_X, saves_X = exact tallies from the match's own event list
```

Possession, pass accuracy/count, shots, corners, and fouls are narrative approximations derived from the same quality gap that produced the scoreline — they don't independently sway it. Yellow cards, red cards, and saves are not approximations: they're counted directly from the events that were actually generated for that match, so the stat line and the event feed always agree exactly.

10. Tournament Awards

Once a champion is crowned, the results screen tallies every match in the whole tournament — every team, not just the viewer's — into a headline summary (matches played, total and average goals, biggest-margin result) plus four awards: Top Scorer, Top Assists, Most Saves, and Player of the Tournament. All purely narrative — none of it feeds back into any score or rating.

10.1 Player of the Tournament

A holistic score across each player's whole tournament, not just their single best stat — goals and assists carry the most weight (as in real Golden Ball voting), saves count for less so one busy goalkeeper in a small sample of matches doesn't automatically dominate, cards are a discipline penalty, and reaching the final with the champion nation is a modest bonus for team success:

```
score = goals × 4 + assists × 2.5 + saves × 0.5
    - yellowCards × 0.5 - redCards × 3
    + (onChampionTeam ? 3 : 0)
```

The highest-scoring player across the entire tournament wins the award; ties are broken by whoever is found first in iteration order (extremely rare in practice given the continuous scoring).

Appendix: Data Disclaimer

The 48-team list matches the real, official field for the 2026 FIFA World Cup, and every single one of the 1,248 players in the game is a real member of that nation's actual 26-man 2026 World Cup squad (sourced from Wikipedia's squad lists, themselves drawn from FIFA's official squad announcements). No fictional or generated players exist anywhere in the dataset — a team with fewer real players in a given position group than a demanding formation calls for simply won't be revealed to a drafter who still needs that position, and a bot fielding that formation with that nation falls back to reusing its weakest already-selected player rather than inventing one.

Overall ratings are fictional gameplay values, not an official or licensed rating dataset. See RULES.md Section 9 for the full breakdown.